Fiscal Year:	FY 2018	**Task Last Updated:**	FY 09/28/2017
PI Name:	Foster, Jamie Ph.D.		
Project Title:	Impact of Microgravity on the Cell-Cell Interactions between a Mutualistic Bacterium and Its Animal Host		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Merritt Island	**State:**	FL
Zip Code:	32953	**Congressional District:**	3
Comments:	EDITOR'S NOTE: Congressional district is for University of Florida, Gainesville. Address is PI's physical address.		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	12/01/2013	**End Date:**	11/30/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2018 per F. Hernandez/ARC (Ed., 11/14/17) NOTE: End date changed to 11/30/2017 per NSSC information (Ed., 9/6/16)		
Key Personnel Changes/Previous PI:	March 2017 - The postdoctoral fellow working on this project received a job and left this project. However, in May 2017 a new graduate student joined the project and will use this research as part of her dissertation research.		
COI Name (Institution):			
Grant/Contract No.:	NNX13AM44G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Beneficial microbes play a significant role in the normal development of animal tissues. Through the formation of healthy stable associations between host and symbiont, particularly during post-embryonic development, carefully regulated developmental pathways can be induced. In this experimental research plan, we propose to examine the impact of microgravity on the normal molecular interactions between host and symbiont. To examine these effects we propose to use the model system between the squid Euprymna scolopes and the luminescent bacterium Vibrio fischeri. For over two decades this squid-vibrio model has served as an ideal ecosystem to examine the influences of mutualistic bacteria on animal development. To simulate microgravity, rotating bioreactors called high-aspect-ratio rotating wall vessel bioreactors (HARVs) will be used to provide a low-shear environment for both the host and symbiont, thereby simulating microgravity. Specifically, we propose the following: Objective 1. Examine the mechanism(s) associated with the delay of bacteria-induced immune cell trafficking that occurs in simulated microgravity. Objective 2. Characterize the changes in the transcriptome of the host squid in the presence and absence of symbiotic bacteria exposure under microgravity conditions. Objective 3. Develop a flight-ready microtiter plate for future flight experiments and optimize the conditions needed to initiate and maintain the symbiosis in this new hardware. The proposed work has several implications for the field of space biology. First, by examining how normal healthy symbioses are initiated and maintained under microgravity conditions, we can begin to determine the sensitivity of these ecosystems to perturbations in the space environment. Secondly, we can assess the genetic and metabolic responses to gravity in the presence and absence of symbiosis, potentially identifying previously unknown triggers or cues associated with post-embryonic animal development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	All animals form symbiotic associations with bacteria. These ubiquitous interactions are essential for the normal development and health of the host organism. This project helps to improve our understanding of the molecular cross-talk that occurs between animals and microbes by delineating those genes and pathways that are influenced by gravity. As the force of gravity has not changed throughout Earth's history we have a unique opportunity to assess how these host-microbe interactions alter gene expression and their physiology in the absence of gravity.
Task Progress:	The results of our YR4 activities can be summarized by the following activities: 1. Peer-reviewed Publications Casaburi, G, I Goncharenko-Foster, AA Duscher and JS Foster (2017) Transcriptomic changes in an animal-bacterial symbiosis under modeled microgravity conditions. Scientific Reports 7:46318 In preparation: Duscher, A., A. Conesa, M. Bishop, and J.S. Foster (2017) Impact of modeled microgravity on the mutualistic bacterium, Vibrio fischeri. Nature Microgravity in preparation. Belcaid, M., Simakov, O., Casaburi, G., Moriano-Guttierrez, S., Suria, A., Pankey, S., Oakley, T., Kremer, N., Koch, E., Goncharenko-Foster, I., Rokhsar, D, McFall-Ngai, M., Foster, J.S., Nyholm, S.V. (2017) The genome and transcriptome of the Hawaiian bobtail squid (Euprymna scolopes): insights into the evolution of symbiosis-specific tissues. In preparation. 2. Oral Presentations Foster, J.S. (2016) Host Transcriptomic Changes in an Animal-Bacterial Symbiosis Under Modeled Microgravity Conditions. American Society for Gravitational and Space Biology, Cleveland, OH Duscher, A.A., J.S. Foster. Impact of modeled microgravity on peptidoglycan recognition receptor protein and immune cascade gene expression in the beneficial symbiosis between the bobtail squid, Euprymna scolopes, and its bioluminescent bacterium, Vibrio fischeri. University of Florida Graduate Research Symposium, Cocoa Beach, FL Simakov, O. D. Rokhsar, M. Belcaid, G. Casaburi, J.S. Foster, S.V. Nyholm (2017) Evolution of complex genome architecture in metazoans, Earlham Institute, Norwich, UK 3. Poster Presentation Vroom, M., A. Troncoso-Garcia, J.S. Foster (2017) Characterizing the bacteria-induced apoptotic pathways in juvenile Euprymna scolopes under simulated microgravity. University of Florida Graduate Research Symposium, Cocoa Beach, FL Casaburi, G., I. Goncharenko-Foster, and J.S. Foster (2017) Transcriptomic changes in an animal-bacterial symbiosis under modeled microgravity conditions. Astrobiology Science Conference, Mesa AZ Duscher, A.A., J.S. Foster (2017) Impact of Modeled Microgravity on the beneficial symbiosis between the bobtail squid, Euprymna scolopes, and its bioluminescence bacterium, Vibrio fischeri. Host Microbe Interactions, University of Southern California, LA, CA Duscher, A. G. Casaburi. M. Bishop and J.S. Foster (2016) Impact of Simulated Microgravity on the Transcriptome of the Mutualistic Bacterium Vibrio fischeri. American Society for Gravitational and Space Biology, Cleveland, OH 4. Research Progress: Task 1. Characterize the changes in the transcriptome of the host squid in the presence and absence of symbiotic bacteria under microgravity conditions. RNAseq - Generating transcriptomic libraries to animal host. We completed this task in 2016 and published the first

Task Progress:	paper on the transcriptome in 2017. However, the transcriptomes under gravity and microgravity conditions have served as a rich resource to begin to look at the molecular mechanisms associated the effects of microgravity on a beneficial symbiosis. A. New peptidoglycan receptor identified in host animal -- Data mining of the host transcriptome has identified a novel peptidoglycan receptor (PRPG) that responds to V. fischeri peptidoglycan, a key molecule for the activation of the host immune system. Although there was a statistically significant increase in the expression of the PRPG in symbiotic animals compared to animals without bacteria, there wasn't a difference between gravity and microgravity conditions at 12 and 24 h. We are in the process of examining earlier time points to assess whether there is differential expression under microgravity conditions. B. Genome assembly of host Euprymna scolopes – In addition to the microgravity studies, the results of this project are being used to generate an assembled transcriptome of the host organism Euprymna scolopes. The transcriptome was assembled earlier this year and has now served as a scaffold to assemble the E. scolopes genome. The genome is now 96% complete, which is the highest for any mollusk, and we are in the process of completing the bioinformatic analyses and preparing this work for publication. Additionally, once the paper has been submitted we will generate an online portal for researchers to access the data and compare the results to the E. scolopes database. Task 2. Examine the mechanism(s) associated with the delay of bacteria-induced immune cell trafficking that occurs in simulated microgravity. Bacterial transcriptomes under simulated microgravity and gravity conditions – This aspect of the project was completed in 2016. However, the results provided some interesting insight into the onset of the apoptotic cell death. A. LPS shedding by V. fischeri is accelerated under simulated microgravity conditions. Our analyses have revealed that under microgravity conditions there is an increase in the shedding of lipopolysaccharide (LPS) from the surface of V. fischeri. LPS is a virulence factor known to induce the immune systems of animals. We are currently characterizing this shedding by studying the production of outer membrane vesicles (OMV) formed due to flagellar activity of the microbes. We are in the process of ascertaining wither the LPS is associated with the OMV by examining several mutants of V. fischeri that have no or hyperflagellar activity during microgravity conditions. B. Elucidating the apoptosis cascade within the bacteria-induced development of the host. Based on the results obtained in Task 1, we are now examining the potential bacterial-induced apoptosis cascade in the host animal and how this might be linked to the immune response. Through bioinformatic data mining and PCR we have not identified the presence of numerous apoptosis genes and are in the process of trying to ascertain if they are differentially expressed under simulated microgravity. Task 3. Delineate the flight hardware for future flight experiments and optimize the conditions needed to initiate and maintain the symbiosis during flight. Techshot Fluid Processing Cassette – We have been working with Dave Reed of Techshot to assess the viability of the host organisms in the Techshot Fluid Processing Cassette (FPC) hardware. We assessed the biocompatibility of the Fluorinated Ethylene Propylene (FEP) bags and the Pharmed BPT tubing as well as the population loading of the bags and found the animals can survive with no additional nutrients or feeding for a total of five days. We are currently working on engineering solution to bring in new seawater and potential nutrients to extend this to a minimum of two weeks.
Bibliography Type:	Description: (Last Updated: 04/20/2026)
Articles in Peer-reviewed Journals	Casaburi G, Goncharenko-Foster I, Duscher AA, Foster JS. "Transcriptomic changes in an animal-bacterial symbiosis under modeled microgravity conditions." Scientific Reports. 2017 Apr 10;7:46318. https://doi.org/10.1038/srep46318 ; PubMed PMID: 28393904; PubMed Central PMCID: PMC5385879 , Apr-2017